

## PALLETPILOT · DESIGN DECISION RECORD

# PalletPilot design for construction

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## DOCUMENT CONTROL

|             |                                                              |         |                |
|-------------|--------------------------------------------------------------|---------|----------------|
| Document ID | PLP-DDR-003                                                  | Version | 0.5            |
| Status      | <b>Draft</b>                                                 | Date    | 2026-10-02     |
| Author      | Amish Chadha                                                 | License | CERN-OHL-S-2.0 |
| Project TRL | <b>TRL 3</b> Proof of concept                                |         |                |
| Repository  | github.com/BoujeeEnjinia1701/palletpilot                     |         |                |
| Source      | docs/decisions/0003-design-for-construction.md @ uncommitted |         |                |

## REVISION HISTORY

| Version | Date       | Author       | Change                                                                                                                                                                                          |
|---------|------------|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0.5     | 2026-10-02 | Amish Chadha | Approved follow-ups carried into the model: 5 mm top plate, tube lever and two hollow pivots (A2); safety laser scanner in place of the lidar (ISO 3691-4); Table 4 added; Consequences updated |
| 0.4     | 2026-10-02 | Amish Chadha | Table 1 changes (P1 to P10) accepted by Amish on 2026-10-02                                                                                                                                     |
| 0.3     | 2026-10-02 | Amish Chadha | A2 and A3 accepted by Amish on 2026-10-02 (R15 prototype limit 45 kg; 10 mm lift for smooth indoor floors only); Table 1 changes still open                                                     |
| 0.2     | 2026-10-01 | Amish Chadha | A1 (steering range) accepted by Amish as recommended, option (a), with the TRL 4 aisle test as a go or no-go gate; Table 1, A2 and A3 stay open                                                 |
| 0.1     | 2026-10-01 | Amish Chadha | Changes that make the concept physically buildable, with the reason for each; made under Amish's 2026-09-30 instruction to make the design physically buildable; open for his review            |

## 0003: Design for construction

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- **Date:** 2026-10-01
- **Status:** accepted. Amish, 2026-10-02: "APPROVED: Design-for-construction changes in 10 repos (CityTwin, CoolShade, PalletPilot, Heliolite, PotholeLog, EarthPress, ReadyKit, CellCheck, CargoMule and ThermaCart)". This covers the changes P1 to P10 in Table 1, made under Amish's 2026-09-30 instruction to make the design physically buildable, and is recorded in the design decisions register (PLP-DEC-001). The items in Table 3 change what the kit does or how it is pitched, so they were proposed separately and listed in the register. A1 is decided: Amish, 2026-10-01: "i accept your recommendations for PalletPilot". He was given the recommendation for A1 only, so A1 is decided as recommended (option a, with the TRL 4 aisle test as a go or no-go gate) and recorded in PLP-DEC-001. A2 and A3 are decided: Amish, 2026-10-02: "i approve your recommendations for all 555 open decisions." A2 is decided as option (a) with R15's prototype limit restated as 45 kg, and A3 as option (a) for smooth indoor floors only; both are recorded in PLP-DEC-001.

### ■ Context

On 2026-09-30 Amish asked for every repo to have an illustrated build plan and wrote: "If you are realising that the design cannot be built as per concept - fix the design assumptions to match and be physically feasible as you draw the illustrations." The TRL 3 model of PalletPilot showed what the kit does, but it was a massing model: several parts had no fixing, floated in space, overlapped other parts, or could not do the job the concept gives them. Checking the model with build123d found the problems in Table 1.

The changes keep what the kit does: the same drive layout (two 200 mm hub motors on a 260 mm track, 180 mm behind the steering axis, 1.5 kN spring preload), the same enclosure under the handle sweep, the same bumper face, lidar scan plane and anchor baseline, the same electrical parts and the same stop chain. The added length (290 mm), handle clearance (40 mm at 70°, 14 mm lowered flat), stopping, traction, energy and tracking figures are unchanged. Every change is in `cad/src/model.py`, which now runs constructability checks (`python cad/src/model.py --check`): 46 pairs of parts that must touch do touch, no two of the 36 components share volume, nothing collides with the release lever raised, and the steering sweep is checked against the jack's frame. All pass.

### ■ Decision

*Table 1. Changes made to the model, the BOM and the calculations.*

| #  | Problem in the concept                                                                                                                                                                            | Change made                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Why this way                                                                                                                                                                                                                                                                                                   |
|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P1 | The subframe was a 20 mm slab with a clamp block 10 mm short of the yoke and a tongue under it, held by nothing. It could not grip the yoke.                                                      | A 6 mm top plate rests on the yoke plate's rear margin with a 62 mm radius notch round the pump. A 10 mm lower jaw goes under the margin, with a packing bar behind the yoke's edge, and four M12 bolts behind the edge pull the jaw and the top plate together, gripping the yoke plate between them.                                                                                                                                                                                        | R1 forbids drilling the jack. The bolts pass behind the yoke's edge, not through it; the pump in the notch stops the plate sliding back, and the packing bar stops it sliding forward.                                                                                                                         |
| P2 | The trailing cheeks were welded solid to the plate, so the wheels could not move, and the two spring towers floated outside the plate touching nothing. The 1.5 kN preload had nothing to act on. | Each hub motor is carried by its own drive arm (10 mm plate) that pivots on a 20 mm pin in two rear hangers behind the wheels. The arm's front end carries a spring tab; a 100 N/mm spring between the tab and a saddle under a cam pushes the arm, and so the wheel, down. The arm ratio of 1.77 gives 750 N per wheel from 423 N per spring.                                                                                                                                                | A pivoted arm is the simplest sprung mount. The pivot behind the wheel leaves the space ahead of the wheel for the spring and the release, and keeps every part between the steer wheels and the bumper. The stiff spring lets the empty jack's steer wheels lift only about 1.2 mm, as PLP-CAL-001 [C3] asks. |
| P3 | The hub motors sat 5 mm from the cheeks with no shaft between them.                                                                                                                               | Each motor's fixed shaft passes through the arm's axle hole with a 5 mm spacer inside and a nut outside.                                                                                                                                                                                                                                                                                                                                                                                      | This is how a single-sided hub servo motor is mounted; the shaft size is confirmed when a motor is chosen.                                                                                                                                                                                                     |
| P4 | The release lever was a bar along the enclosure side with no pivot and no link to the springs, so it could not lift anything.                                                                     | A 16 mm cam shaft runs across under the top plate in two front hangers, with a 22 mm eccentric cam above each spring saddle and a crank on its left end. A link joins the crank to an up-arm on a 400 mm lever pivoted on a bracket on the top plate, making a parallelogram. Raising the lever a quarter turn lifts the saddles 22 mm; slotted straps then lift the arms, and the drive wheels clear the floor by 10 mm. With the wheels down the lever lies forward on a stop, over centre. | One lever lifts both wheels (R14) with a peak effort of about 58 N, half the 117 N the concept estimated [G4]. Every moving part was checked in both lever positions.                                                                                                                                          |
| P5 | The bumper was a single solid block, and its mounts ran from the cheeks (which now move with the wheels) to the hoop's sides, which reached forward to 70 mm behind the steering axis.            | A U hoop of 40 x 20 x 2 mm tube, its side legs bolted with two M8 bolts each into rivet nuts on brackets welded under the top plate. The safety edge is riveted to the front (50 mm deep, 40 mm travel) and to both sides (20 mm). The side legs start 120 mm behind the steering axis.                                                                                                                                                                                                       | The hoop is fixed to the parts that do not spring, so the bumper face stays put. The bumper face, width (480 mm) and edge travel are unchanged, so R8 stands. Edge length is now about 0.9 m, not 1.3 m.                                                                                                       |
| P6 | The lidar bracket hung from the subframe in the space the arm pivot now needs.                                                                                                                    | A bent steel strap screwed to the top of the hoop's front bar carries the lidar shelf.                                                                                                                                                                                                                                                                                                                                                                                                        | Scan plane (200 mm) and the lidar's position 10 mm behind the bumper face are unchanged.                                                                                                                                                                                                                       |
| P7 | The two bumper anchors floated above the bumper with no fixing.                                                                                                                                   | Each anchor sits on a small corner plate on top of the hoop's front corners, its top 180 mm above the floor, below the scan plane.                                                                                                                                                                                                                                                                                                                                                            | Baseline (450 mm) unchanged, so the tracking budget [U1] to [U4] stands.                                                                                                                                                                                                                                       |
| P8 | The enclosure had no fixing; the rear emergency                                                                                                                                                   | Four aluminium angle feet riveted to the box sides, each bolted through the top plate with an M8 bolt;                                                                                                                                                                                                                                                                                                                                                                                        | The box floor stays whole, so nothing inside sits on a bolt head.                                                                                                                                                                                                                                              |

| #   | Problem in the concept                                                                                                                                                                                              | Change made                                                                                                                                                                   | Why this way                                                                                                                                                                             |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     | stop and the disconnect overlapped its walls.                                                                                                                                                                       | holes in the walls for the stop (22 mm) and the disconnect (40 mm).                                                                                                           | The box is 89 mm deep plus a 6 mm lid, 95 mm as before.                                                                                                                                  |
| P9  | With the kit on, the yoke cannot turn as far as a bare jack: the drive end reaches the jack's frame head. On the reference donor the top plate's corners met it at 34° and the cams, link and wheels at 44° to 48°. | The top plate's front corners are cut off at 45° (50 mm), and two rubber steering stops under them meet the frame head flat at 40° each way, 4° before any other kit part.    | Without stops, steel parts of the kit would strike the jack's frame. The stop angle is set by the donor survey (R1). The range itself is a change to what the kit does: see Table 3, A1. |
| P10 | The donor reference geometry overlapped itself: the steer wheels ran through the yoke plate and the frame head, and the pump through the yoke.                                                                      | Reference donor corrected: yoke plate 204 to 219 mm above the floor over 180 mm steer wheels, frame head's rear face 100 mm ahead of the steering axis, yoke side legs added. | Reference geometry only; still a survey item (R1, O3).                                                                                                                                   |

Table 2. Knock-on changes.

| Item         | Change                                                                                                                                                                                                                                                                              | Reason                                                  |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| Mass         | Kit 44.4 kg (was 42.0 kg) [B1]: drive module 10.6 kg (was 9.8), release mechanism 3.3 kg (was 0.8), bumper 2.3 kg (was 3.3), steering stops 0.1 kg. R15 mass is now 4.4 kg over its 40 kg target (2.0 kg was accepted for now in PLP-DDR-002).                                      | Made parts are now weighed from their modelled volumes. |
| Cost         | BOM lines 1 (\$90 to \$130), 3 (\$45 to \$50) and 13 (\$20 to \$45) repriced and line 19 (steering stops, \$10) added: \$1,690. Value-engineering target: USD 1,610 ( budget_usd , unchanged). Estimated cost of the constructable design: USD 1,690 (USD 80 over the target) [K1]. | Parts added for construction.                           |
| Calculations | PLP-CAL-001 v0.4: mass, loads and forces rerun (design total 1,108 kg; worst motor torque 25.6 N·m of 30; ramp margin 1.53; bumper-only stop 39 mm of 40 mm; 28 % of usable energy left); lever effort [G4] 58 N; steering range [G5]. No requirement changed status.               | Follows the model.                                      |
| Drawings     | PLP-DWG-001 Rev P4; making sketches PLP-DWG-101 to 111 added; concept blueprint PLP-DWG-010 Rev P2.                                                                                                                                                                                 | Follows the model.                                      |
| Documents    | PLP-REQ-001 v0.6 and PLP-PRC-001 v0.6: mass, cost, release and steering figures.                                                                                                                                                                                                    | Follows the model.                                      |

Table 3. Proposed; A1 accepted by Amish as recommended on 2026-10-01, A2 and A3 on 2026-10-02.

| #  | Question                                                                                                                                                                                                                                           | Options                                                                                                                                                                                                                                                                                                                                               | Recommendation                                                                                                                                                                                                                                                         |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A1 | Steering range. With the kit fitted the jack steers about 40° each way, against about 90° bare, on the reference donor. The concept says the kit "steers with the handle like the bare jack"; it does not. A wider turn needs more room in aisles. | (a) Accept for the prototype, set the stops from the donor survey and measure the turning circle at TRL 4; (b) narrow the drive end (hoop sides inboard, release moved) for perhaps 48°, at the cost of a narrower anchor baseline and a worse R5; (c) a different drive layout, for example one centre drive wheel under the yoke, which reopens D3. | (a), and state the steering range in the precis and README; make the TRL 4 aisle test a go or no-go gate: if the kit cannot make a right-angle turn into a standard pallet bay, change the drive layout (option c) before TRL 5. <b>Accepted by Amish, 2026-10-01.</b> |
| A2 | Mass. The kit is 44.4 kg against R15's 40 kg; Amish accepted 2.0 kg over for now.                                                                                                                                                                  | (a) Accept for the prototype and recheck at the motor quote; (b) lighten now: 5 mm top plate, tube lever, hollow pivot pin (about 1.5 kg).                                                                                                                                                                                                            | (a), keeping (b) as savings to try in the TRL 4 drawings. <b>Accepted by Amish, 2026-10-02</b> , with R15's prototype limit restated as 45 kg and rechecked at the motor quote.                                                                                        |
| A3 | Wheel lift. The lever lifts the drive wheels 10 mm clear. A larger lift needs a larger cam, which the top plate's height does not leave room for.                                                                                                  | (a) Accept 10 mm for smooth indoor floors; (b) lower the cam shaft and raise the hangers for about 15 mm.                                                                                                                                                                                                                                             | (a). <b>Accepted by Amish, 2026-10-02</b> , for smooth indoor floors only; the build plan says so.                                                                                                                                                                     |

Table 4. Follow-ups of the 2026-10-02 decisions, carried into the model.

| #  | Change                                                                                                                                                                                                                                                                                                                                        | Why                                                                                                                                                                                                                       |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| F1 | The top plate is 5 mm (was 6 mm), the release lever is a 20 x 10 x 2 mm steel tube (was a solid bar), and the single 20 mm pivot pin is replaced by two hollow 20 mm tubes with a 4 mm wall, one per drive arm, each held by an M12 bolt through its bore. The enclosure sits 1 mm lower on the thinner plate and is 96 mm deep.              | A2 (approved 2026-10-02): about 2.0 kg saved. Two short pivots also leave the middle of the frame free for the scanner.                                                                                                   |
| F2 | The lidar is replaced by a safety laser scanner (SICK nanoScan3 class, about 107 x 80 x 118 mm) on a 3 mm steel shelf on the hoop's front bar, scan plane 173 mm above the floor (was 200 mm). The safety edge is 68 mm high (was 80 mm) and the anchor tops are 165 mm (was 180 mm) so that both stay clear of the shelf and the scan plane. | Follow mode is treated under ISO 3691-4 (approved 2026-10-02), which needs a safety-rated personnel sensor. The body is larger than the lidar's, and it must fit between the bumper and the enclosure with 5 mm to spare. |
| F3 | Personnel stop: scanner to the safety relay, monitored controlled stop (stop category 1) by the driver, brakes close at standstill; the relay opens both contactors if the monitored time runs out. Field 0.80 m (was 0.86 m); the delay is 0.15 s (was 0.25 s).                                                                              | PLP-CAL-001 v0.7 [S4], [S5], [S5a].                                                                                                                                                                                       |

The constructability checks now include clearances for the scanner and lever (nine gaps). The steering stops, the donor survey items (packing bar thickness, clamp holes, notch, stop angle) and everything else in Tables 1 to 3 are unchanged; the survey items wait for the three surveyed jacks.

## ■ Consequences

- `design_state: constructable` in `project.yaml`. The build plan PLP-BLD-001 shows every component and step in pictures generated from the model ( `cad/src/build_plan_media.py` ); open decisions are in PLP-DEC-001.
- Requirement status is unchanged by the changes: 10 met, 2 not met (R5, R15 mass), 2 at risk (R7, R9), 3 not verifiable at TRL 3 (R1, R6, R16). With A2 decided on 2026-10-02 (R15 prototype limit 45 kg) it is 11 met and 1 not met (R5); with the follow-ups of Table 4 the kit is 42.9 kg and R17 is reported against the value-engineering target, USD 3,455 over (USD 5,065 against USD 1,610; USD 80 over before the safety scanner).
- The appearance model `cad/src/product_model.py` now takes its drive module, release, hoop, scanner shelf and steering stops from the constructable model (2026-10-02), and the render scenes are exported. The photoreal renders ( `media/render-*.png` ), `media/card.png` and `media/social-preview.png` still show the earlier design until they are remade on Amish's Mac, where Blender is.
- With A1 accepted (2026-10-01), the prototype steers about 40° each way with the kit fitted, set by rubber stops (a bare jack turns about 90°), for a turning radius of roughly 1.5 m about the load wheels; the range is confirmed in the TRL 4 aisle test. The precis (PLP-PRC-001 v0.7), README, requirements (PLP-REQ-001 v0.7) and build plan (PLP-BLD-001 v0.2) state this range. The TRL 4 aisle test is a go or no-go gate: if the kit cannot make a right-angle turn into a standard pallet bay, the drive layout changes before TRL 5 (option c, for example one centre drive wheel under the yoke, which reopens D3 of PLP-DDR-001).
- The donor survey (R1, O3) now sets four numbers in the build: the yoke plate's thickness (packing bar), its rear edge and the pump's position (notch and clamp), and the frame head's position (steering stop angle).